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A comprehensive techno-economic review of microinverters for Building Integrated Photovoltaics (BIPV)

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ABSTRACT

Increasing incentives for building integrated photovoltaic (BIPV) generation at local/major grid levels established it as a viable decentralized option promising large growth potential. BIPV systems – comprising photovoltaic (PV) modules installed on available building surfaces and downstream energy conversion devices – afford improvement of energy efficiency of buildings and partially or fully dispenses with the supply of centralized grid power. This paper presents a comprehensive techno-economic review, covering the technical as well as commercial aspects of microinverter technology. Advantages of microinverters over conventional inverters are detailed along with a discussion on economics of its installation in distributed solar generation systems. Different power converter topologies reported in the available literature are presented. The paper also reports the historical development of commercial microinverters and their present status in the Photovoltaic (PV) market. Survey of the available products from some of the technology leaders in the market has been done and their specifications are tabulated. The paper concludes with a discussion on the necessities of the next generation microinverters for increased penetration in the PV market.

Contents

1. Introduction

Increasing energy demand coupled with rapid depletion of limited fossil fuel reserves has led to the increase in global requirement for clean energy sources and efficient and reliable

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harnessing technologies [1–5]. Compared to fossil fuel based systems, technologies for utilization of renewable energy resources like Solar, Wind, Geothermal, Biomass, Fuel Cell, Ocean, Tidal, etc. are environmentally clean as these leave minimal deleterious footprints, the reason why such sources are more relevant and important in this century. Solar energy in its primary and secondary forms is at the forefront as an effective, inexhaustible and environmental friendly source of energy [6–10] and utilization technologies include solar heating, solar photovoltaic and solar thermal electricity. Moreover, energy generation from PV modules is among the most reliable and constitutes a simple and elegant method of harnessing solar energy as it is clean, pollution free and modular [11]. PV modules also enjoy a host of advantages like absence of moving parts, noiseless operation, low maintenance cost, operational simplicity, small footprint, compact area and commercial abundance. The cost of PV modules, initially contributing significantly to the total system cost has also come down at present owing to substantial increase in production volumes and rapid growth in semiconductor technologies, leading to installation of large megawatt scale power plants integrated with mainstream power grids [12,13]. Solar power plants require large areas for installation in locations promising availability of abundant solar energy, and ideal large scale PV installations are remote from cities and are located in deserts, large tracts of barren lands, etc. [14]. Long transmission lines are therefore required to connect large solar power plants located in remote areas to the load centers. While the output of these plants can complement the conventional power generation in meeting the load demands, there was a felt necessity to focus on the techniques for harnessing solar energy in populated urban and rural areas. This led to the concept of Building Integrated Photovoltaics (BIPV) through the integration of PV in the buildings wherein Photovoltaic (PV) materials and systems are used to replace conventional building materials and systems in part of buildings [15,16] so as to simultaneously serve as building envelope material and power generator. In recent years, the solar industry has seen radical change in the inverter space. On the utility side, inverters are getting bigger in order to service massive, multi-megawatt power plants. On the residential side or Building Integrated Photovoltaics (BIPV), inverters are getting smaller and more adaptable, opening up the market for "plug and play" systems. This has created more choices for consumers and installers. Miniaturized voltage converters – generically called Microinverters – attached to the back of individual solar panels converts DC power generated by the solar panel into AC power compatible with the voltage level of the local utility grid. Microinverters simplify set-up and power conversion in solar energy conversion systems and render the systems more efficient, reliable and safe. In conventional solar PV systems, panels are strung together in a series and are linked to a single centralized power inverter. One major drawback of this integration is system-wide failure, much like old-fashioned festival lights, where when one panel fails, the entire system fails. Microinverters enable individual panels to operate independently, greatly increasing the system efficiency and also eliminate the need for additional wiring, complex parts and large, noisy, heat-producing centralized inverters as proposed in [17]. Optimal energy extraction is achieved by using maximum power point tracking algorithm, which enables fast and efficient tracking of the maximum power point at all levels of solar irradiation, rapidly changing shading conditions and low light environments and ensures that available energy is harvested to the maximum extent from the panel at all times [18].

This point onwards the paper is organized as follows: Section 2 discusses conventional inverter configurations for BIPV solutions; Section 3 establishes the claim of microinverters as a viable decentralized option for BIPV; Section 4 discusses the power converter topological configurations available in the literature; Section 5 dwells on the status of microinverters in the PV market; Section 6 highlights the requirements for next generation of microinverters and Section 7 carries the conclusion.

2. Conventional inverter configurations for BIPV solutions

BIPV systems can provide savings in materials and electricity costs and improve the architectural and esthetic facets to buildings due to their sleek look, variety of shapes, colors and forms, and the ability of these materials to blend in perfectly with the backdrop ambience. BIPV solutions are increasingly being incorporated in the construction of new buildings as a principal or ancillary source of electrical power [19]. While BIPVs can be both stand-alone and grid connected, the latter is gaining ascendancy because of its increased reliability, thereby adding to benefits accruing from a cooperative utility policy where both the building owner and the utility grid controllers stand to gain [20–23]. Grid connected BIPV also entails a lower system cost due to the absence of storage elements like battery banks [24,25]. The power electronic interface for grid connected BIPV systems performs two major tasks: (a) it converts the dc voltage generated by a PV module(s) to the grid compatible ac voltage level, and (b) it operates the photovoltaic module at its Maximum Power Point (MPP) for maximizing energy capture from the PV system. These tasks must be performed with highest possible efficiency over a wide power range. The MPP is tracked by means of a Maximum Power Point Tracking (MPPT) algorithm. The current injected into a single-phase grid has to follow a sinusoidal pattern in phase with the grid voltage in keeping with the norms specified by IEEE standards 519 and 1992 [26], which specifies that the maximum allowable amount of injected current harmonics for system short circuit ratio (SCR) should be less than 20, and the total harmonic distortion (THD) in the voltage level less than 5% [27].

Different inverter configurations are possible for BIPV; details pertaining to the common ones are provided here:

a. *Centralized configuration*: previously the primary technology used for dc–ac conversion was the 'centralized' inverter for BIPV system interconnected with the grid, where a number of PV modules were connected in series and parallel arrays to obtain the required power and voltage levels [28] (Fig. 1(a)). This configuration has severe drawbacks since high voltage dc cabling is required to connect the BIPV system to the centralized inverter, which creates problems of safety, cable losses, dc arcs and fire hazards. Centralized MPPT performed on the solar array as an aggregate leads to less than optimal power harvest because of inefficient tracking of the maximum power point resulting from shadowing circumstances of individual panels in the system, consequent module mismatch and the resultant complex current–voltage curves. Minimum number of BIPV panels required to set up the system is high so as to reach the required voltage levels, as there is no provision for voltage boost in this system. Space requirement of centralized inverter also increases with increase in the power rating, warranting the provision of costly cooling systems [29–31]. Expandability is also limited, as is the flexibility in voltage, current and power levels once the system is designed and set up.
b. *String configuration*: to overcome some of the disadvantages of centralized inverters, string inverters, as shown in Fig. 1(b), have been developed [32]. In this configuration, a group of series-connected BIPV modules called 'strings' have their own inverters; the voltage generated by individual strings is high enough to attain the grid voltage level. The losses connected with poor MPPT are less in this configuration while the reliability and modularity is improved in comparison to

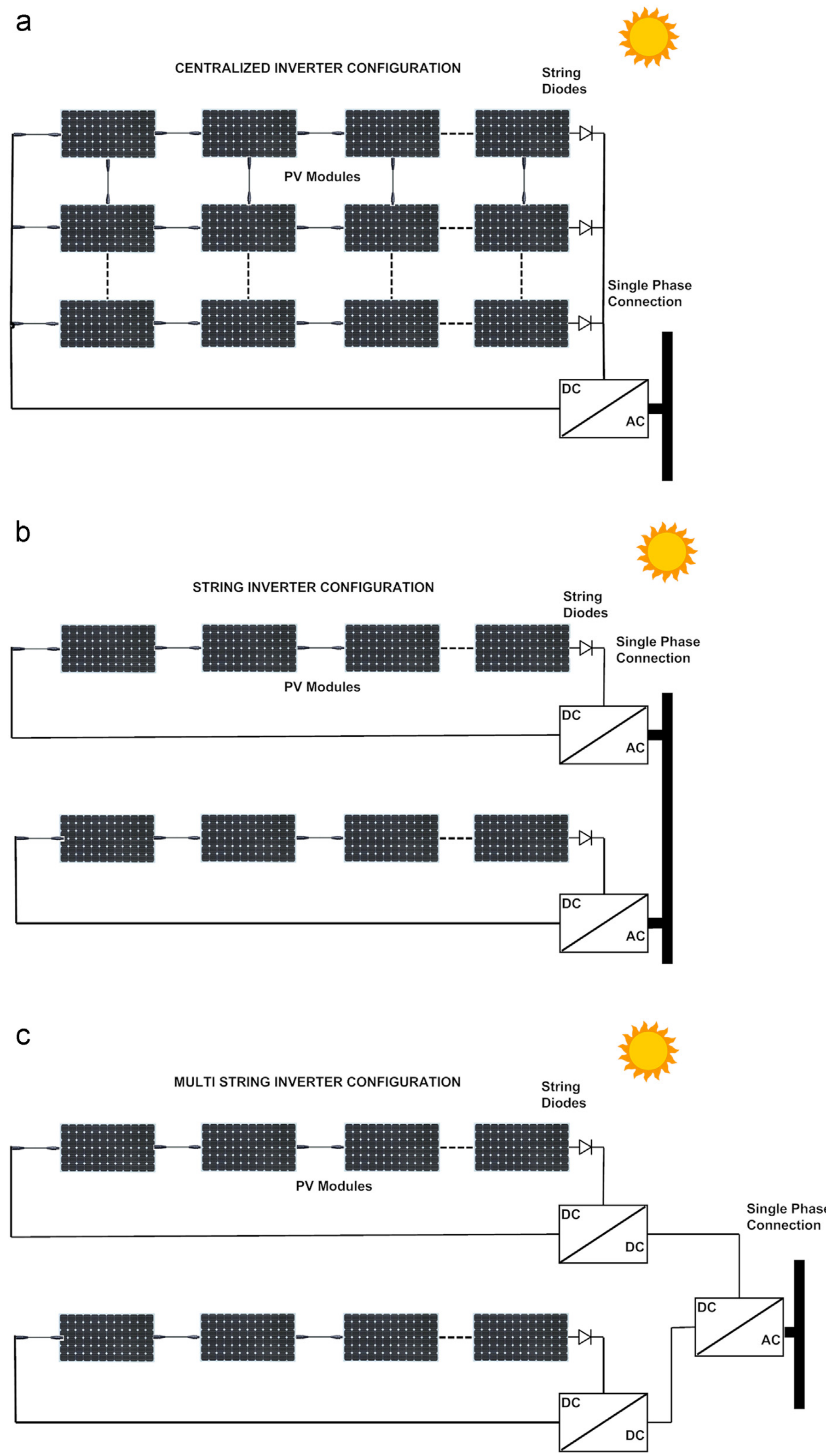

Fig. 1. (a) Centralized Inverter Configuration (b): String Inverter Configuration (c): Multi-String Inverter Configuration.

centralized inverters. However, problems like complex dc cabling and associated issues remain even in this configuration. In addition, like their centralized counterparts, string inverters require auxiliary cooling and protection devices. The necessity of deploying semiconductor circuit elements of higher ratings and the resultant system bulkness are issues that do not act in favor of string inverters.

c. *Multi-string configuration*: multi-string inverters constitute an improvement over string inverters. In the multi-string configuration, every string has its own dc–dc converter, which individually connects to a centralized dc–ac inverter (Fig. 1 (c)). Multi-string inverters enjoy all the advantages of a string inverter system; additionally, dc–dc converters dedicated to each string impart more flexibility to the system as compared to the string inverters since the provision for voltage boost is present in this configuration [33].

3. Microinverter: an elegant solution for BIPV

Unlike centralized, string or multi-string configurations that aggregate and convert power generated by arrays of BIPV modules, the microinverter configuration (Fig. 2) evolved as a means for improving upon the shortcomings associated with the aforementioned inverter configurations [34]. Microinverters take the concept of string inverters to the next level – providing dc–ac conversion from individual panels rather than an entire string. A microinverter converts dc power generated from a single PV module to ac power i.e. every module has its own integrated power electronic interface to connect it to the utility grid [28,29].

3.1. Advantages of microinverters for a BIPV system

Microinverters have the following enumerated advantages as compared to traditional systems in BIPV [35]:

1. *Simplicity* – use of ac cabling and connectors instead of dc reduces installation costs and simplifies system design. Microinverter architectures also enable simpler wiring, which translates to lower installation costs [36].
2. *Intelligence* – simple but accurate monitoring down to the level of an individual PV module; the ability to track energy produced on a per-PV-module basis comes in handy for monitoring and diagnosing PV module related issues that are more difficult in series/parallel PV arrays [37].
3. *Failure detection* – to secure economical profits for PV systems, cost-effective methods that detect system failures as quickly as possible are of paramount importance to increase the operating efficiency of a PV system [38].
4. *Reliability* – the system is rendered more reliable, smart, and efficient through module level optimal MPPT by energy harvest [39].
5. No mismatch losses at system level as every AC-module operates at its individual MPP.
6. No need for bypass and string diodes.
7. Removal of problems due to lightning-induced surge voltages in the PV module substrings.
8. Lesser dc voltage compared to central inverters, rendering the system less prone to arcing and leading to enhanced safety.
9. Better payback on the initial investment [40].
10. Low minimum system size of one AC-module, lowering the threshold for individuals to start their own PV plant.

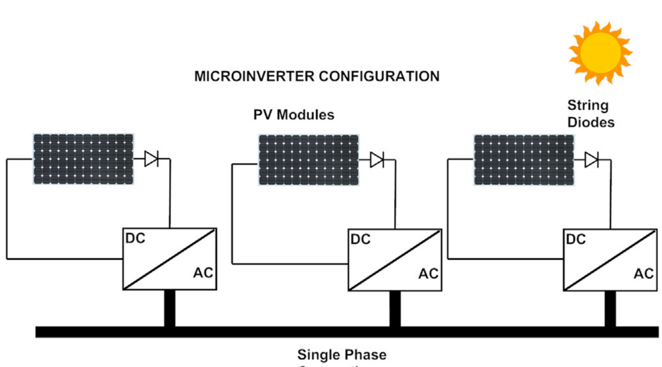

Fig. 2. Microinverter Configuration.

All these features make ac units an interesting option for the integration of photovoltaic systems into buildings. Independent operation of ac units is particularly important, since this prevents the poor functionality of a PV system due to partial shading of one or more PV modules. Existing literature suggests that in case of partial shading, ac units outperform string units and results in higher annual yields well above 5% [41,42]. The reason for higher yield is the absence of mismatch losses common in traditional PV systems. With both centralized and string inverter architecture, the shading on one module affects the entire string. A report submitted by the consultants to the California Energy Commission states that in a sample of 119 PV systems, 85% experienced some shading, with 30% having more than 5% reduction in output due to shading whereas the average reduction in power over the sample size was of the order of 7%. Shading in itself poses a serious threat to power output in residential and small commercial installations, and the ability of microinverters to reduce the deleterious effects of shading by isolating shaded modules is among the strongest and best-verified selling points of microinverter technology [43].

3.2. Economics of microinverter installations in BIPV

The cost involved in a typical PV installation comprises capital and recurring costs. Capital costs include the cost of solar modules, cost of the balance of systems and other installation costs. Cost of solar modules remains same for all the configurations. Cost of microinverter modules put together may be slightly higher than a single centralized inverter or multiple string inverters [44]. But installation costs are less for microinverters because of absence of high voltage dc switch boxes and protection circuits that need specialist skills and training to install [45]. Recurring costs include maintenance and repair costs. Solar PV systems with central or string inverters are relatively expensive to maintain as the monitoring can only be done at a system level and not at the level of each individual module which renders fault detection difficult, time consuming and expensive. In contrast, ac units can be monitored individually even online through a web-enabled interface – thereby facilitating the exact location of faults [37]. Achieving life time at par with PV units is a difficult task in string inverters, which is reflected in limited warranties of perhaps 5–10 years. Microinverters, in contrast, are widely expected to achieve a life time of 15–20 years in the near future [46]. This information is important from the consumer's point of view and is an essential factor to consider while deciding on the investment in a PV system; lesser the life time more are the costs of replacement of these expensive components and more is the cost of labor involved. White paper published by Enecsys microinverters quantifies these numbers based on their field experience with a 7 kWp BIPV system [45], which are given in the Table 1.

In a comparison of 25 kWp BIPV systems deployed using Solarex MST-50MV tandem junction thin film silicon modules (in the form of fully integrated module and a frame less laminate), Posbic et al. [47] found that microinverter systems are measurably less expensive than central inverters (Table 2). The 25 kW volume represents about 10 central inverters versus over 200 microinverters.

This comparison can however hold good only at lower volumes. With large volume systems (of several MWp) advantages of AC

Table 1
Cost of 7 kWp BIPV system from the experience of Enecsys microinverters.

Sl.	*Name of the head*	*Microinverter* ($)	*String inverter* ($)
1.	Electronic module costs	3500	5250
2.	Installation costs	12,500	16,500
3.	Maintenance costs	739	5192
Total		**16,739**	**26,942**

Table 2
Cost of 25 kWp BIPV system from the experience of installation with Solarex MST-50 MV thin film modules.

Sl.	Name of the head	Roof integrated (laminated module) $/Wp DC		Roof stand off (framed module) $/Wp DC	
		Central inverter	Micro-inverter	Central inverter	Micro-inverter
1.	Modules	3.00	3.00	3.60	3.60
2.	Balance of system (BOS) and installation	2.86	2.57	2.13	1.92
3.	Inverter accessories	1.71	1.43	1.71	1.43
	Total	**7.57**	**7.00**	**7.44**	**6.94**

module inverters begin to reduce and centralized inverters prove to be the better option [47].

4. Power conversion topological configurations of microinverters

Panel voltage for the popular mono crystalline and multi-crystalline silicon PV modules is around 25–40 V dc at maximum power point operation [28]. Main challenge of a microinverter is to step up this PV panel voltage to the range of utility supply voltage (which is 120–230 V ac). One option is to use a line-frequency transformer that affords a high step-up gain through the selection of a proper turns-ratio. However, bulk and high leakage inductance restricts this option for practical applications [28,48].

Next best option reported in the literature suggests the use of dc–dc boost converter to obtain the required amplification of the voltage which can be achieved by using transformer or transformerless topologies. High frequency transformer based topologies have simple structure and are easy to control as the amplification is achieved by selecting appropriate turns ratio [48]. Also high frequency transformers allow for dual grounding on both input and output terminals thereby avoiding resonance between stray capacitances of the PV panel and the inductances in the current main paths [28]. However high frequency transformers can reduce the efficiency of system and increases the input current ripple from the PV panel. Reduction of current ripple warrants the use of electrolytic capacitors resulting in decrease of life time and an increase in overall system size and cost [49,50]. To eliminate this problem some topologies with current fed schemes are proposed [49]. Transformer less conventional boost converters are to be operated at extremely low duty cycles to achieve high amplifications required in a microinverter. However as the switch turn off period becomes short ripple in the currents increase thereby increasing the conduction loss and voltage stresses of the power semiconductor switches [50]. Several options like power device paralleling, multilevel boost converters, cascaded boost converters, interleaved boost converters, coupled inductor and switched capacitor circuits are proposed in the literature to obtain the required boost [50–52]. Some of the transformer based topologies report the use of pseudo dc link in which case the dc–dc converter stage provides the fully rectified sine modulated output instead of the constant dc voltage. Because of this, power electronic switches in the dc–ac inverter stage can be operated only at the grid frequency, thereby improving the efficiency. But the control of this topology is quite complex as the output needs to be rectified and modulated for a 100 Hz sinusoidal wave [48].

A novel dc–ac and ac–ac cascaded approach, in which system a dc–ac high frequency inverter provides the voltage amplification along with sinusoidal modulation and an ac–ac inverter converts the high frequency current to a low frequency current for grid synchronization reported in some of the literatures. It has advantages in terms of low component count and high efficiency, but the construction of bidirectional switches adds complexity to the design [48]. Fig. 3 provides different topological configurations of microinverter technology in the existing literature.

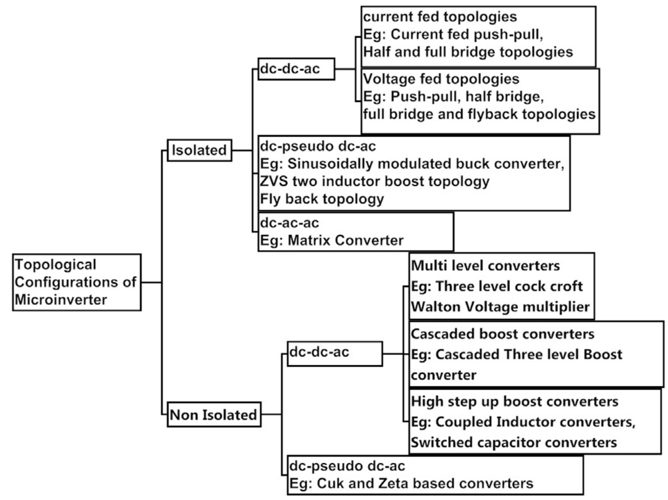

Fig. 3. Different topological Configurations of Microinverter.

Instantaneous power injected from the inverter into the grid is a level shifted sinusoid whose average power is equal to the input dc power. This requires power decoupling between the PV panel and the grid to buffer and release the energy instantaneously upon requirement. Many of the available topologies in the literature make use of capacitors as the decoupling element. Though electrolytic capacitors find widespread use, these are less reliable and sensitive to temperature variations [53–55]. To eliminate the use of bulky electrolytic capacitors some of the topologies report use of active power decoupling techniques where in decoupling is achieved by use of small capacitors in tandem with power semiconductor switches [56–59]. But these techniques may increase the cost and complexity of the system and reduce the efficiency.

5. Penetration of microinverters in the PV market

Development of microinverters started in late eighties at with the institutes like Institute for Solar Energy Distribution Technology (ISET), Germany and Ascension technology, USA and Zentrum fuer Sonnenenergie-und Wasserst off-Forschung ZSW, Germany pioneering the research in related technologies. ZSW, Germany came up with the first prototype of a 50 W Microinverter in 1992. Ascension Technology, AES was the first US Company to commercially develop microinverters (Sun Sine 300 W) [60], which was introduced during 1996–97. It was done in collaboration with ASE 300 W and 15 kW installation was also done at pentagon, 1999 [60].

Many of the microinverters were optimized for specific PV modules. For example, the Sunmaster-130S was developed for Shell 100 W modules (24 V), the Sunsine 300 for the ASE 300 W module (36 V) and the MI250 for the Solarex 240 W module (48 V). Other earlier attempts include the OK4 100 W inverter developed by OKE for an NKF 72 cell PV-module (1994, Netherlands), the Mastervolt 130 W inverter of Mastervolt for a 72 cell PV module (1994, Netherlands), SunSine 300 W inverter of Ascension Technology for a 270 W ASE module (1996, USA), AES 250 W transformerless inverter of SDA (Stephen Strong & Robert Wills) for a 144 cells PV module (1996, USA) and PV2GO, 130 W inverter for a 72 cells PV-module (2000, Europe) [61,62]. Microinverters subsequently disappeared from the market because these early attempts did not have the requisite efficiency, reliability or economy to address mainstream solar energy needs, where higher voltages, higher efficiency and lower costs are essential [60–62].

A growing market for solar energy and the availability of better and more efficient electronic components have now triggered off the advent of new and improved microinverters for solar energy applications. According to IMS Research's report on the market development and market demand for microinverters, the total global market share for microinverters and power optimizers was 250 MW in 2010, which represents 1.4% of the global installed capacity for PV generation. Total revenue for the industry was estimated to be $100 million [43]. Use of these technologies in the PV inverter industry grew strongly in 2010 with shipment of microinverters growing by well over 500% even though they still accounted for less than 1% of PV inverter revenues, whereas string and central inverters accounted for 46% and 53% respectively [63]. Microinverters also continued to be the fastest growing segment of the PV inverter market, growing by 180% in 2011 with revenues surging by 160% to more than $200 million, according to a 2012 report from IMS Research. In 2012 the forecast projected a growth in excess of 70%, reaching nearly 900 MW [64]. Further, the prospect for microinverters remains bright with a report envisaging more than 10% penetration into the market in 2016 amounting to revenues of nearly $1.5 billion. Some of the biggest and attractive markets will be the United States, Canada, United Kingdom, Australia and Japan, wherein microinverters can capture a share of 25% in the total PV market by 2016 [64].

Enphase Energy – who pioneered the development of microinverters in 2008 – was the major player in the microinverter market in 2011 with a share of 90% of shipments. In 2011 the number of microinverter units shipped by Enphase Energy reached 5,00,000 [65]. Enecsys, U.K. came up with microinverters enjoying a service life of over 25 years, matching that of solar modules. Enecsys products were also warranted for 20 years over an ambient temperature range of $-40\,^{\circ}$C to $+85\,^{\circ}$C. Solar Bridge Technologies (USA) launched its microinverter in 2010, and is active in establishing partnerships with leading PV module manufacturers to integrate the developed microinverters with PV modules [66]. They also offer a 25-year warranty on their products, which is comparable with the life of PV modules to which they are integrated. This is notably greater than the warranties of 20 and 15 years offered by Enecsys and Enphase respectively. Direct Grid Technologies [67], came up with Closed Loop MOSFET Planar technology with remarkable improvement in both thermal management and electrical efficiency.

Product survey revealed that there were no less than 26 microinverters available commercially. Tables 3(a) through 3(c) provide a comparison of these microinverters in terms of chosen parameters. Source data was collated from the web pages, brochures and data sheets available from the individual manufacturers.

6. Requirements of next generation microinverters

Though promising tremendous growth potential, there are certain technical and commercial issues that the next generation microinverters should properly address. Technical requisites like efficiency and reliability and commercial issues like cost, market visibility play a vital role in deciding the future of penetration of microinverters in the PV market. In this section, we deal with some such salient issues.

6.1. Increased reliability and life

High reliability is a major operational requirement for microinverters; their average life time should also match the standard warranty of 20–25 years that accompany most of the PV modules. This is particularly important as the microinverters are often mounted on roof tops and in locations that are difficult to access, thereby increasing the costs associated with frequent maintenance. The life time of a microinverter is also an issue of importance in calculating the life cycle cost of electricity.

Electronic components in a microinverter system experience higher thermal stresses because of the following reasons [68]:

1. Ambient temperatures can vary from 80 °C at noon under maximum irradiation to 0 °C during the night, especially in desert regions leading to huge thermal stresses.
2. Internal temperatures of microinverters can be 25–30 °C higher than the ambient temperatures as they are mounted on the roof top at the back of panels.
3. Self heating of microinverter due to internal losses further enhances this temperature.

A major challenge for the next generation microinverters therefore is the capacity to withstand these cyclic thermal stresses and achieve a life time of 25 years or more. Some of the switching devices used in inverters, relays and electrolytic capacitors are also the potential weak points for inverter reliability. Electrolytic capacitors are very sensitive to temperature where an increase in 10 °C can halve the lifetime of the electrolytic capacitor due to loss of electrolyte [28,53–55]. The next generation microinverters should therefore explore the use of film capacitors which are rugged, can withstand higher voltage surges and have self-healing properties inherent in their construction [28]. In addition, new designs should be tried out involving active power decoupling techniques. Reliability can further be enhanced by limiting the electrical and thermal stress on components through the proper thermal management design.

6.2. Increased efficiency

The efficiency of microinverters has to be high enough to compete with classical string or central inverters. Achieving efficiencies of about 95% may be a difficult task because of the lower power levels involved and also because of the fact that each microinverter will have its own signal processing circuitry, protection circuitry with relays, etc. that contribute to increase in losses in comparison to centralized inverters. The high frequency transformers provided for the galvanic isolation between the panels and grid for compliance with the leakage current standard add to the losses of microinverters [69]. Choice of the power converter topology plays a vital role in the success of the product and may affect the key parameters, cost, overall size, EMC performance and efficiency. Cascaded topologies have lower efficiencies because of the losses involved in two stages of the converter. When hard-switching techniques are used, the switching loss tends to be high as the semiconductors in both conversion stages switch under high frequencies. Soft-switching technique can be utilized in both conversion stages to minimize the drawback of this arrangement, but that may result in a higher component count and therefore a higher cost and a lower reliability [51]. The use of single stage topologies with pseudo-dc link can, however, reduce the losses to some extent as the second stage needs to be switched only at power frequencies.

Semiconductor switches have a major influence on the converter efficiency. Inspite of unique characteristics like higher break down voltage, reduced recovery voltage applicability of SiC (Silicon Carbide) [70] its applicability in microinverters may be limited as the module voltages and currents are very low. However they do find applications on the grid connection side because of relatively high voltages involved. GaN-on-Si (Gallium Nitride on Silicon) technology developed by International Rectifier Corporation (IR)

Table 3
(a) Commercial microinverter models and comparison of parameters and features.

MICROINVERTER MAKE	*PV power* (W)	*MPPT voltage* (V)	*Max./min voltage* (V)	*Max. DC current* (A)	*Short circuit current* (A)	*Night time power consumption* (mW)	*Output power (Max.)* (W)
APTRONIC	360	20–50	60/18	11	No information	30	330
DIRECT GRID S250	250	145–165	190	2.7	No information	10	250
DIRECT GRID S400	400	145–165	190	2.7	No information	10	400
DIRECT GRID S460	460	53–63	75	10	No information	10	460
DIRECT GRID S460 (GEMINI)	530	53–65	75	10	No information	No information	530
ENECSYS D360W-72	360	30–42	54/27	13.4	16	< 30	360
ENECSYS D480W-60	480	24–35	44/20	22	32	< 30	450
ENECSYS S200W-72	200	30–42	54/27	7.4	16	< 30	185
ENECSYS S240W-60	240	23–35	44/20	12	16	< 30	225
ENECSYS S217W-60	240	23–35	44/20	12	16	< 30	217
ENECSYS S263	280	30–42	54/27	10.4	16	< 30	263
ENPHASE D380	230	22–40	56/28	10	12	50	380
ENPHASE M190	230	20–40	56/28	10	12	< 30	190
ENPHASE M210	240	31–50	62/32	10	12	< 30	210
ENPHASE M215	190–260	22–36	45/22	10.5	15	46	215
SUNSHINE 250 NA	320	20–50	60/20	10	No information	0	300
I ENERGY GT 260	240	30–50	59/25	10	12	< 30	265
SOLAR BRIDGEP 235 LV 240	250	18–36	48	11.25	14	< 30	225
SPARQS 190W	205	32–50	65/33	10	No information	< 26	190
SPARQS 215W	230	32–50	65/33	10	No information	< 25	215

(b)

MICROINVERTER MAKE	*Output power (Min.)* (W)	*Output current (Max.)* (A)	*Power factor*	*Harmonic distortion* (%)	*Peak inverter efficiency* (%)	*Euro weighted efficiency* (%)	*Normal MPPT tracking efficiency* (%)
APTRONIC	No information	1.6	> 0.99	No information	93	91.1	99.8
DIRECT GRID S250	238	1.14	> 0.995	No information	95	92	99
DIRECT GRID S400	372	1.63	> 0.995	No information	94	92	99
DIRECT GRID S460	430	2	> 0.99	No information	93	92	99
DIRECT GRID S460 (GEMINI)	416	2	> 0.99	No information	93.7	93.4	99
ENECSYS D360W-72	No information	1.48	> 0.95	< 5	95.4	93.5	No information
ENECSYS D480W-60	No information	1.96	> 0.95	< 5	95	93.5	No information
ENECSYS S200W-72	No information	0.8	> 0.95	< 5	94	91	No information
ENECSYS S240W-60	No information	0.98	> 0.95	< 5	94	91.5	No information
ENECSYS S217W-60	No information	0.9	> 0.95	< 5	94.8	93.5	No information
ENECSYS S263	No information	1.12	> 0.95	< 5	95.4	No information	94.5
ENPHASE D380	No information	1.8	> 0.95	No information	95	95.5	99
ENPHASE M190	No information	0.92	> 0.95	No information	95.5	95	99.6
ENPHASE M210	No information	1	> 0.95	No information	96	95.5	99.6
ENPHASE M215	No information	1	> 0.95	No information	96.3	96	No information
SUNSHINE 250 NA	No information	1.5	> 0.95	< 3	94.8	93.5	> 99.5
I ENERGY GT 260	240	0.88	> 0.95	< 3	94	93	99.3
SOLAR BRIDGE P 235 LV 240	No information	0.93	> 0.99	< 5	95.5	94.5	99.6
SPARQS 190W	No information	0.8	> 0.99	No information	95	93.4	No information
SPARQS 215W	No information	0.9	> 0.99	No information	95	93.4	No information

(c)

MICROINVERTER MAKE	*Operating temperature* (°c)	*Dimensions (mm × mm × mm)*	*Weight* (kg)	*Warranty* (years)	*Enclosure rating*
APTRONIC	−25 to +70	314 × 267 × 66.5	2.5	No information	IP65
DIRECT GRID S250	−40 to +65	225 × 227.5 × 3	2.177	20 (limited)	NEMA-4, IPX4
DIRECT GRID S400	−40 to +60	225 × 227.5 × 3	2.177	20 (limited)	NEMA-4, IPX4
DIRECT GRID S460	−40 to +65	225 × 227.5 × 3	2.177	20 (limited)	NEMA-4, IPX4
DIRECT GRID S460 (GEMINI)	−40 to +65	225 × 233 × 3.175	2.177	20 (limited)	NEMA-4, IP54
ENECSYS D360W-72	−40 to +85	262 × 160 × 35	1.8	20	IP66
ENECSYS D480W-60	−40 to +85	262 × 160 × 40	2.1	20	IP66
ENECSYS S200W-72	−40 to +85	262 × 160 × 35	1.8	20	IP66
ENECSYS S240W-60	−40 to +85	262 × 160 × 35	1.8	20	IP66
ENECSYS S217W-60	−40 to +85	262 × 160 × 35	1.8	20	UL 50 Type 4 ×
ENECSYS S263	−40 to +100	262 × 160 × 35	1.8	20	UL 50 Type 4 ×
ENPHASE D380	−40 to +65	311 × 152.4 × 33.8	2.83	15	NEMA 6
ENPHASE M190	−40 to +65	203.2 × 133.35 × 31.75	2	15	NEMA 6
ENPHASE M210	−40 to +65	203.2 × 133.35 × 31.75	2	15	NEMA 6
ENPHASE M215	−40 to +65	173 × 164 × 25	1.6	25	NEMA 6
SUNSHINE 250 NA	−40 to +60	260 × 200 × 45	2	25	IP65
I ENERGY GT 260	−40 to +65	232 × 201 × 43.1	1.6	15–25	IP66
SOLAR BRIDGE P 235 LV 240	−40 to +65	203.2 × 165.1 × 31.75	2.04	25	NEMA 6
SPARQS 190W	−40 to +65	190.5 × 127 × 30.48	1.95	25	NEMA 6
SPARQS 215W	−40 to +65	190.5 × 127 × 30.48	1.95	25	NEMA 6

enable dramatic improvements of power converter efficiencies as the GaN transistor has high mobility of electrons (HEMT) there by making the product (RDS(on) × Qsw) 33% less than the last generation Si MOSFET. Also the characteristic is “soft” due to the essential absence of holes thereby eliminating the minority carrier effect in reverse recovery charge Qrr resulting in the elimination of otherwise needed filtering (snubber) circuitry [71].

6.3. Reduced cost

Combined cost of microinverters is greater than that of a string inverter or centralized inverter for a particular BIPV installation. Lux Research estimates that the cost of microinverters averages between USD 0.50/watt and USD 1.00/watt [43]. In case of central and string inverters, these are typically below USD 0.30/watt and USD 0.50/watt respectively. Higher costs of microinverters can therefore limit the market penetration of microinverters and the cost therefore has to ideally match that of the centralized option.

6.4. Reactive power management

Next generation of microinverters can incorporate the important grid stabilization feature of reactive power management [72]. In Europe, PV systems above 30 kW are required to vary the power factor between 0.9 and 1. This means that if microinverters are used in PV systems larger than 30 kW, reactive power compensation should be done at the level of the individual microinverter or the same has to be achieved through a central compensation unit. A power factor of 0.9 requires over-dimensioning of the microinverter's hardware by nearly 11% [73], and adds to the cost, which should be brought down to an acceptable level.

6.5. Application specific integrated circuit (ASIC) for controllers

Next generation of microinverters can be based on ASIC (application-specific integrated circuit) technology which will help in greatly reducing the number of components and associated cost while increasing the reliability and service life even further [74].

6.6. Visibility of product and benefits

Finally, the adoption of microinverters require good visibility and market education as to the potential benefits these can offer as in many cases system owners, developers and installers choose to stay with less expensive options that they are familiar with [43]. Some microinverter companies have chosen to partner with established module and inverter makers, which is expected to assist both visibility and market education [43,66].

7. Conclusion

This paper has provided a comprehensive insight into the concept of microinverters, specifically in relation to the emerging application of BIPV. Advantages of microinverter technology like simplicity, reliability, safety and intelligence in comparison with conventional inverter configurations establish them as a viable option for distributed power generation in BIPV. Existing literature suggests that in case of partial shading, microinverter units outperform string units and results in higher annual yields well above 5%. Economics of installation has also been discussed where in the cost per unit for installation of microinverter systems is less than conventional inverter configurations in low power distributed generation systems. Though the cost of electronic modules is more, it is the installation cost of BOS and other accessories which make the difference in overall cost of the system. According to the experience of Enecsys microinverters, while the electronic module costs of microinverter are approximately 1.5 times lower than that of string inverter, installation costs and maintenance costs for string inverter are higher by 1.3 times and 7 times respectively. Tracing the development of the microinverter from the eighties' decade of the last century to the current stage, the paper argues in favor of microinverter deployment for the emerging PV market and forecasts increase in penetration of microinverters in the PV market up to 10% with countries like United States, Canada, United Kingdom, Australia and Japan, expected to have a share of 25% in the total PV market by 2016. Different topological configurations of Power converter reported in the literature were discussed in considerable detail. The paper carries a comprehensive comparison in terms of chosen parameters of different microinverters available commercially whose power range varies from 200–530 W. Peak efficiencies are in the range of 93–96%, while Euro weighted efficiencies are in the range of 91–96%. Weight of the microinverters is in the range of 2–2.5 kg and many of them provided warranty in the range of 15–20 years. Finally, the paper draws attention to the technical and associated requirements necessary for the next generation microinverters and provides significant leads. Reliability and life time have to be improved to make next generation microinverters competitive in the solar photovoltaic market, by employing cutting edge technologies and employing alternatives to less reliable components like electrolytic capacitors, mechanical relays etc. Alternative topologies using new generation semiconductor devices have to be employed for increasing efficiency and thereby make an impact on payback time.

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References

[1] Nakicenovic N, Grubler A, McDonald A. Global energy perspectives. New York: Cambridge University Press; 1998.
[2] Ekins P. Step changes for decarbonising the energy system: research needs for renewables, energy efficiency and nuclear power. Energy Policy 2004;32(17):1891–904.
[3] Olah GA, Goeppert A, Prakash GKS. Beyond oil and gas: the methanol economy. 2nd ed.. Weinheim, Germany: Wiley; 2006.
[4] Key World Energy Statistics. International energy agency, ⟨http://www.iea.org/publications/freepublications/publication/kwes.pdf⟩; 2012 [accessed 24.07.13].
[5] Herzog AV, Lipman TE, Kammen DM Renewable Energy Sources, ⟨http://rael.berkeley.edu/sites/default/files/old-site-files/2001/Herzog-Lipman-Kammen-RenewableEnergy-2001.pdf⟩; [accessed 24.07.13].
[6] Ghosh AK. Solar cells: present and future technology and cost. In: Meyer RF, Olson JC, editors. Long term energy resources. Montreal: Pitman; 1981. p. 761–71.
[7] Goetzberger A, Hebling C, Schock HW. Photovoltaic materials, history, status and outlook. Mater Sci Eng 2003;40(1):1–46.
[8] Wohlgemuth JH, Frederick MD, Cunningham DW, Monus P, Miller J. Long term reliability of photovoltaic modules. In: Proceedings of the conference record of the 2006 IEEE 4th world conference on photovoltaic energy conversion; 2006. p. 2050 –2053, 2.
[9] Kazmerski LL. Solar photovoltaics R&D at the tipping point: a 2005 technology overview. J Electron Spectrosc Relat Phenom 2006;150(2-3):105–35.
[10] Costello D, Rappaport P. The technological and economic development of photovoltaics. Annu Rev Energy 1980;5(1):335–56.
[11] Nema P, Nema RK, Rangnekar S. A current and future state of art development of hybrid energy system using wind and PV-solar: a review. Renew Sustain Energy Rev 2009;13(8):2096–103.
[12] Denholm P, Margolis RM. Very large scale deployment of grid-connected solar photovoltaics in the united states: challenges and opportunities. United States: National Renewable Energy Laboratory; 2006.
[13] Kurtz SR Opportunities and challenges for development of a mature concentrating photovoltaic power industry. Technical report NREL/TP-520–43208.

National Renewable Energy Laboratory. p. 1–16, ⟨http://www.nrel.gov/docs/fy13osti/43208.pdf⟩; 2007 [accessed 24.07.13].
[14] Turney D, Fthenakis V. Environmental impacts from the installation and operation of large-scale solar power plants. Renew Sustain Energy Rev 2011;15(6):3261–70.
[15] Daniel FM, Pieterjan V, Joanna C Building Integrated Photovoltaics: an overview of the existing products and their fields of application. European Photovoltaic Industry Association, ⟨http://www.cagbctoronto.org/files/BIPV_Overview_existing_products.pdf⟩; [accessed 24.07.13].
[16] Spencer S, Dan G Residential grid-tied photovoltaic systems. Energy business intelligence, ⟨http://www.flintenergies.com/documents/ResidentialGridTiePhotovoltaicSystems.pdf⟩; [accesse 24.07.13].
[17] ⟨http://www.sparqsys.com/aboutmicro.html⟩; [accessed 24.07.13].
[18] Esram T, Chapman PL. Comparison of photovoltaic array maximum power point tracking techniques. IEEE Trans Energy Convers 2007;22(2):439–49.
[19] Steven S Building Integrated Photovoltaics (BIPV), ⟨http://www.wbdg.org/resources/bipv.php⟩; [accessed 24.07.13].
[20] Ruther R, Dacoregio MM. Performance assessment of a 2kWp grid-connected, building-integrated, amorphous silicon solar photovoltaic installation in Brazil. Prog Photovolt: Res Appl 2000;8:257–66.
[21] Jardim CDS, et al. The strategic siting and the roofing area requirements of building-integrated photovoltaic solar energy generators in urban areas in Brazil. Energy Build 2008;40(3):365–70.
[22] Yu GJ, So JH, Jung YS, Kang GH, Choi JY Performance results of 15KW BIPV sunshade system. In: Proceedings of the conference record of 31th IEEE photovoltaic specialists conference; 2005. p. 1722–1725.
[23] Prasad DK, Snow M. Designing with solar power: a source book for building integrated photovoltaics (BiPV). London: Images Publishing Group Pty Limited and Earthscan; 2005.
[24] Lewis M. ASHRAE (American Society of Heating, Refrigerating and Air-Conditioning Engineers, Inc). Building for the future is the supplement journal 2004;46(9):S22–9.
[25] So JH, Jung YS, Yu GJ, Choi JY, Cho JH. Performance results and analysis of 3 kW grid-connected PV systems. J Renew Energy 2007;32(11):1858–72.
[26] IEEE recommended practices and requirements for harmonic control in electrical power systems, IEEE Std 519-1992, New York, NY, USA: Institute of Electrical and Electronics Engineers, Inc., 1993.
[27] Daniel JC, Thomas JD, Thomas MB Price and performance considerations for harmonic solutions, ⟨http://www.newark.com/pdfs/techarticles/eaton/Eaton_Technical_Articles/Power_Quality_White_Papers/Harmonic_Solutions.pdf⟩; [accessed 24.07.13].
[28] Kjaer SB, Pedersen JK, Blaabjerg F. A review of single-phase grid-connected inverters for photovoltaic modules. IEEE Trans Ind Appl 2005;41(5):1292–306.
[29] Kjaer SB,Pedersen JK, Blaabjerg F Power inverter topologies for photovoltaic modules—a review. In: Proceedings of the conference record of 37th IAS annual meeting industry applications conference; 2002. p. 782–788, 2.
[30] Meinhardt M, Cramer G Past, present and future of grid connected photovoltaic- and hybrid-power-systems. In: Proceedings of the IEEE power engineering society summer meeting; 2000. p. 1283–1288, 2.
[31] Markvart T. Solar electricity. 2nd ed.. New York: Wiley; 2000.
[32] Calais M, Myrzik J, Spooner T, Agelidis VG Inverters for single-phase grid connected photovoltaic systems – an overview. In: Proceedings of 33rd IEEE power electronics specialists conference; 2002. p. 1995–2000, 3.
[33] Meinhardt M, Cramer G, Burger B, Zacharias P. Multi-string-converter with reduced specific costs and enhanced functionality. Sol Energy 2001;69(6):217–27.
[34] Balfour JR, Shaw M, Bremer N. Advanced photovoltaic system design. Burlington, Massachussets: Jones & Bartlett Publishers; 2011.
[35] Hayman AK. Development of a High-efficiency Solar Micro-inverter. MA: Massachusetts Institute of Technology, Department of Electrical Engineering and Computer Science; 2009.
[36] Ogboenyiya K Taking Energy Efficiency to the Next Level, Texas instruments white paper, SPRAAZ6A, ⟨http://www.ti.com/lit/wp/spraaz6a/spraaz6a.pdf⟩; May 2009 [accessed 24.07.13].
[37] Problems and disadvantages in current residential & commercial on-grid P.V. systems. Solar edge technologies, ⟨http://ussolarinstitute.com/wp-content/uploads/2012/11/se-problems-and disadvantages-in-current-residential-and-commercial-on-grid-pv-systems.pdf⟩; [accessed 24.07.13].
[38] Drews A, et al. Monitoring and remote failure detection of grid-connected PV systems based on satellite observations. Sol Energy 2007;81:548–64.
[39] Enslin JHR, Wolf MS, Snyman DB, Swiegrs W. Integrated photovoltaic maximum power point tracking converter. IEEE Trans Ind Electron 1997;44(6):769–73.
[40] Enslin JHR, Snyman DB. Combined low-cost, high-efficient inverter, peak power tracker and regulator for PV applications. IEEE Trans Power Electron 1991;6(1):73–82.
[41] Oldenkamp H et al. PV-wirefree versus conventional PV-systems: detailed analysis of difference in energy yield between series and parallel connected PV-modules, ⟨http://www.ecn.nl/docs/library/report/2004/rx04074.pdf⟩; [accessed 24.07.13].
[42] Gross MA, Martin SO, Pearsall NM Estimation of output enhancement of a partially shaded BIPV array by the use of AC modules. In: Proceedings of the conference record of 26th IEEE photovoltaic specialists conference; 1997. p. 1381–1384.
[43] Christian R Microinverters and power optimizers – perspectives of distributed PV system architecture in the residential market, ⟨http://www.solarserver.com/solar-magazine/solar-report/solar-report/microinverters-and-power-optimizers-perspectives-of-distributed-pv-system-architecture-in-the-residential-market.html⟩; [accessed 24.07.13].
[44] Maehlum MA Microinverters V/S Central Inverters, ⟨http://energyinformative.org/are-solar-micro-inverters-better-than-central-inverters/⟩; [accessed 24.07.13].
[45] Microinverter Economics in Solar PV installations, ⟨http://tindosolar.com.au/assets/pdfs/MicroInverterEconomics.pdf⟩; [accessed 24.07.13].
[46] Larsen K Microinverters vs. string inverters for solar PV, ⟨http://www.renewableenergyfocus.com/view/10472/micro-inverters-vs-string-inverters-for-solar-pv/⟩; 2010 [accessed 29.07.13].
[47] Posbic JP, Rever III WB. Economic optimization of building integrated photovoltaic systems. In: Proceedings of the 1998 international solar energy conference, June, Albuquerque, NM, USA; 1998. p. 209–212.
[48] Li Q, Wolfs P. A review of single phase photovoltaic module integrated converter topologies with three dc link configurations. IEEE Trans Power Electron 2008;23(3):1320–33.
[49] Liu B, Liang C, Duan S Design considerations and topology selection for DC-module based building integrated photovoltaic system. In: Proceedings of the 3rd IEEE conference on industrial electronics and applications; 2008 p. 1066–1070.
[50] Li W, He X. A review of non-isolated high step up DC/DC converters in photovoltaic grid connected applications. IEEE Trans Ind Electron 2011;58(4):1239–50.
[51] Caro JCR et al. A family of DC–DC multiplier converters. Engineering letters, ⟨http://www.engineeringletters.com/issues_v19/issue_1/EL_19_1_10.pdf⟩; 2011 [accessed 24.07.13].
[52] Lima TDS MPPT for a Photovoltaic Micro-Inverter. Project report written in the scoop of integrated master in electrical and computer engineering, ⟨http://paginas.fe.up.pt/~ee04186/ftp/tese_ee04186_final_v1.pdf⟩; 2012 [accessed 24.07.13].
[53] Erickson RW, Rogers AP A microinverter for building –integrated photovotlaics. In: Proceedings of the twenty-fourth annual IEEE applied power electronics conference and exposition; 2009. p. 911–917.
[54] Rodriguez C, Amaratunga GAJ. Long – life time power inverter for photovoltaic ac modules. IEEE Trans Ind Electron 2008;55(3):2593–601.
[55] Bower W, West R, Dickerson A Innovative PV microinverter topology eliminates electrolytic capacitors for longer life time. In: Proceedings of the conference record of the IEEE 4th world conference on photovoltaic energy conversion; 2006. p. 2038–2041, 2.
[56] Harb S, Haibing H, Kutkut N, Batarseh I, Shen ZJ A three-port photovoltaic (PV) micro-inverter with power decoupling capability. In: Proceedings of the twenty-sixth IEEE annual applied power electronics conference and exposition; 2011. p. 203–208.
[57] Hu H, et al. A three port fly back for pv microinverter applications with power pulsation decoupling capability. IEEE Trans Power Electron 2012;27(9):3953–64.
[58] Hu H, Harb S, Kutkut N, Batarseh I, Shen ZJ. a review of power decoupling techniques for microinverters with three different decoupling capacitor locations in PV systems. IEEE Trans Power Electron 2013;28(6):2711–26.
[59] Ming CY, Yao LC. Three-port flyback-type single-phase micro-inverter with active power decoupling circuit. Phoenix, AZ: IEEE Energy Conversion Congress and Exposition; 2011, 17–22 September p. 501–506.
[60] Katz D Micro-inverters and AC modules – then next generation solar energy solution or An idea from the past, ⟨http://av.conferencearchives.com/pdfs/091001/33.1308.pdf⟩; [accessed 24.07.13].
[61] Oldenkamp H, Jong IJD A.C. modules: past, present and future. Workshop installing the solar solution, Hatfield, UK, ⟨http://www.oke-services.nl/downloads/199801_acmodules_paper_iss1998.pdf⟩; 22–23 January 1998 [accessed 24.07.13].
[62] Oldenkamp H, Jong IJD The return of the AC-Module Inverter. The Netherlands, section 5, ⟨http://www.oke-services.nl/ok4all/ok4all.htm⟩; 2009 [accessed 24.07.13].
[63] Microinverters & power optimizers grow 500% in 2010, but still a long way to mainstream 2011, ⟨http://www.imsresearch.com/press- release/Microinverters_Power_Optimizers_Grow_500_in_2010_But_Still_a_Long_Way_to_Mainstream⟩; [accessed 24.07.13].
[64] Microinverters & power optimizers outpace pv inverters in 2011–70 percent growth predicted in 2012, ⟨http://www.imsresearch.com/press-release/Microinverters_Power_Optimizers_Outpace_PV_Inverters_in_2011_70_percent_Growth_Predicted_in_2012⟩; [accessed 24.07. 13].
[65] Wesoff E Enphase PV Microinverter milestone: half million units shipped, ⟨http://www.greentechmedia.com/articles/read/enphase-hits-half-million-units-shipped-milestone⟩; [accessed 24.07.13].
[66] ⟨http://solarbridgetech.com/solarbridge-technologies-partners-with-auo-to-launch-ac-modules-to-north-american-market/⟩; [accessed 24.07. 13].
[67] ⟨http://www.directgrid.com/corporate/company-profile/⟩; [accessed 24.07.13].
[68] Russell MC The promise of reliable inverters for PV systems: the microinverter solution. Green ray solar, ⟨http://www.greentechmedia.com/articles/read/the-promise-of-reliable-inverters-for-pv-systems-the-micro-inverter-solutio⟩; 18 June 2010 [accessed 24.07.13].
[69] Ho CNM, Challenges and design considerations of pv inverters in the future smart grids. IET, ⟨http://carlho88.webs.com/Paper/Conf/2012/PV_inv.pdf⟩; [accessed 24.07.13].
[70] Sugawara Y, et al. 12-19 kV 4H–SiC pin diodes with low power loss. In: Proceedings of 13th international symposium on power devices and ICs 001. p. 27–30.

[71] Šoja M, Ikić M, Banjanin M, Radmanović MD. Improving efficiency of power electronics converters. Electronics 2010;14(2):37–42.
[72] Nasser K An AC PV module with reactive power capability: need and benefit. Petra Solar Inc., ⟨http://www.harvesterenergycanada.com/pdf/VAR-Capability.pdf⟩; [accessed 29.07.13].
[73] Stalter O, Kranzer D, Rogalla S, Burger B Advanced solar power electronics. In: Proceedings of the 22nd international symposium on power semiconductor devices & ICs; 2010. p. 3–10.
[74] ⟨http://www.frost.com/prod/servlet/press-release.pag?docid=266236645⟩; [accessed 24.0713].